

Review Questions

Hydrogen, Batteries and Fuel Cells

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Hydrogen

H1. Give three ways to produce green hydrogen in a fossil fuel free manner.

Answer:

1. Renewable electrolysis using electricity from photovoltaics, wind, hydro, or geothermal energy.
2. Photoelectrochemical water splitting using sunlight.
3. Solar thermal water splitting using high-temperature solar energy.

H2. Why is hydrogen considered an energy carrier rather than a primary energy source? How does its role differ from that of fossil fuels?

Answer:

Hydrogen is considered an energy carrier rather than a primary energy source because it is generally bound in water or hydrocarbons, and there are no large naturally occurring reservoirs of free hydrogen on Earth that can be extracted directly. Therefore, electrical, thermal, or other forms of energy must first be supplied to produce hydrogen. The produced hydrogen stores this input energy and can be transported to the point of use, where the energy can be released through fuel cells or combustion.

In contrast, fossil fuels are naturally occurring primary energy resources that can be extracted directly.

H3. Describe how hydrogen can be stored and transported.

Answer:

Storage

1. **Physical-based storage:** compressed gaseous hydrogen, cryo-compressed hydrogen, and liquid hydrogen.
2. **Material-based storage:** adsorption in porous materials and storage in hydrogen carriers, including metal hydrides, ammonia, chemical hydrides, and liquid organic hydrogen carriers (LOHCs).

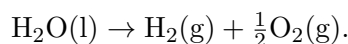
Transportation

1. Pressurized gaseous hydrogen can be transported through pipelines or in high-pressure containers.
2. Liquid hydrogen requires cryogenic conditions and insulated storage tanks.
3. Hydrogen can also be transported in a chemical precursor form.

H4. From a thermodynamic perspective, what is the minimum theoretical energy consumption for water electrolysis?

Answer:

The overall reaction for water electrolysis is



The minimum reversible electrical work equals the Gibbs free-energy change. At 25 °C under standard-state conditions,

$$\Delta G = 237.16 \text{ kJ mol}^{-1}, \quad M_{\text{H}_2} = 2.016 \times 10^{-3} \text{ kg mol}^{-1}.$$

Hence, the minimum specific electrical energy is calculated from

$$\begin{aligned} w_{\min} &= \frac{\Delta G}{M_{\text{H}_2}} \\ &= \frac{237.16}{(2.016 \times 10^{-3})(3600)} \text{ kWh kg}^{-1} \\ &\approx 32.677 \text{ kWh kg}^{-1}. \end{aligned}$$

This is the minimum electrical input, with heat supplied to the reversible process. Practical electricity consumption is higher, about 50–55 kWh kg⁻¹.

H5. What is the “hydrogen economy”? What technological and infrastructural bottlenecks must be overcome to realize it?

Answer:

The hydrogen economy is a future energy system in which hydrogen serves as an important energy carrier and gradually replaces part of the current use of fossil fuels. It connects renewable power generation, hydrogen production, storage and transportation, hydrogen refueling stations, fuel cells, and end-use applications.

Technological bottlenecks include:

- Low-cost, efficient, and low-carbon large-scale hydrogen production.
- The low volumetric energy density of hydrogen.
- The cost, durability, and service life of fuel cells.
- Safety issues such as hydrogen leakage, flammability, and hydrogen embrittlement.

Infrastructural bottlenecks include:

- Large-scale renewable power generation and electrolyzers.
- Hydrogen pipelines and transportation networks.
- Regional and point-of-use storage facilities.

H6. In certain proportions of hydrogen and oxygen an explosive gas is achieved. What fractions are needed?

Answer:

Hydrogen : Oxygen = 2 : 1. This corresponds to approximately 66.7% hydrogen and 33.3% oxygen by volume.

Batteries

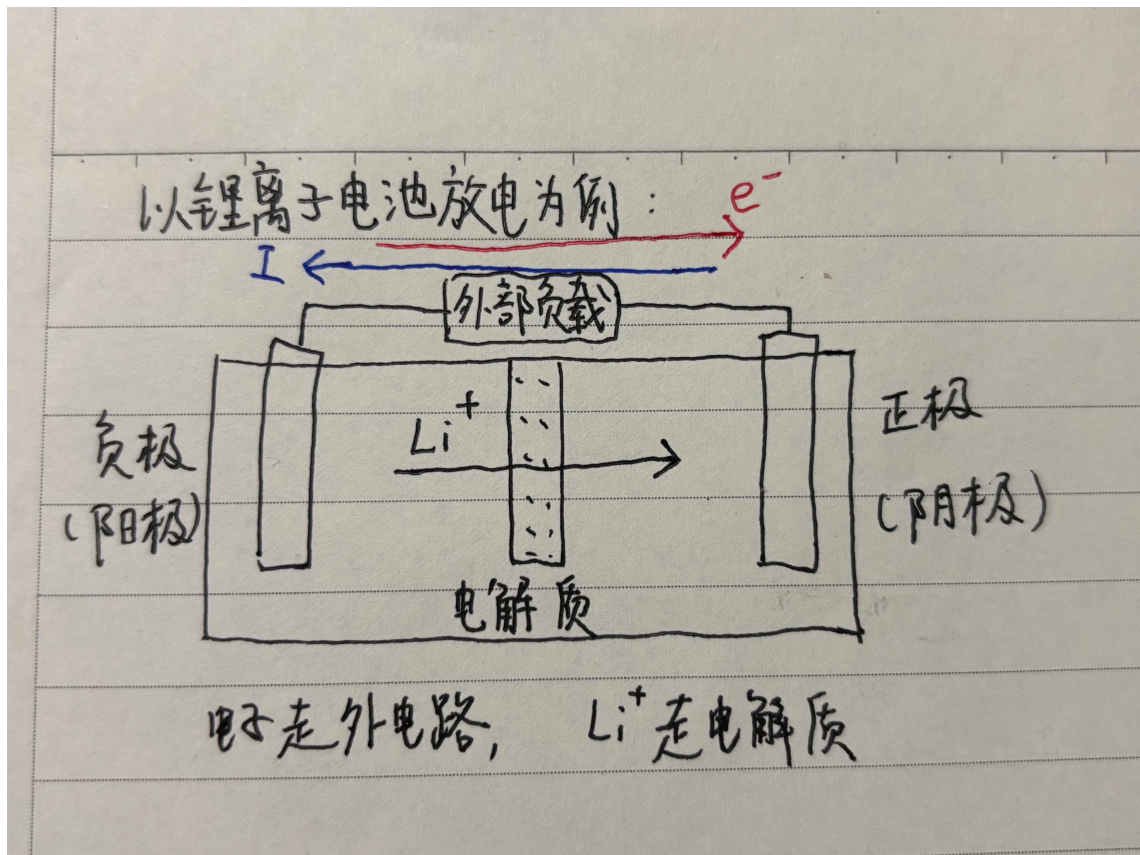
B1. Define or tell what is meant by:

- Gibbs energy.** A thermodynamic potential defined as $G = H - TS$. At constant temperature and pressure, $\Delta G < 0$ indicates a spontaneous process.
- Electrolytic cell.** A cell that uses electrical energy to drive a non-spontaneous chemical reaction, as during battery charging.
- Galvanic cell.** A cell that produces electrical energy from a spontaneous chemical reaction, as during battery discharging.
- Battery capacity and its units.** The total charge stored or delivered by a battery. $Q = I \times t$; units: Ah or mAh.

- e. Battery watt-hours.** An estimate of the energy content of a battery. $E = V \times Q$; units: Wh or kWh.
- f. Energy density of a battery.** Energy stored per unit volume, in Wh/L. Energy per unit mass is the specific energy, in Wh/kg.
- g. SOC.** State of charge, the ratio of available capacity to nominal capacity: $\text{SOC} = (Q_{\text{available}}/Q_{\text{nominal}}) \times 100\%$.
- h. DoD.** Depth of discharge, the percentage of the battery capacity that has been used: $\text{DoD} = 100\% - \text{SOC}$.
- i. Half cell.** One electrode and its associated electrolyte in a battery, representing either an oxidation or reduction half-reaction.
- j. Redox.** Reduction-oxidation reaction; oxidation releases electrons and reduction accepts electrons.
- k. SOH.** State of health, an indicator of remaining capacity relative to the beginning of life. It also reflects power fade and changes in internal resistance.
- l. Degradation.** The deterioration of battery performance with time or cycling, including capacity loss, power fade, and increased internal resistance.
- m. BTMS.** Battery thermal management system. It controls cell temperature and temperature uniformity and helps prevent thermal runaway.
- n. BMS.** Battery management system. It monitors and manages the battery's power, energy and state functions, including SOC, SOH, and temperature.
- o. C-rate.** The charge or discharge rate of a battery relative to its nominal capacity. A 1C rate means the battery is charged or discharged in one hour, 0.5C in two hours, and 2C in half an hour.

B2. Show a principal sketch of the current/electron flow and movement of ions during discharge of a battery.

Answer:



B3. Describe the cell voltage versus current density under load. Identify the various losses or overpotentials.

Answer:

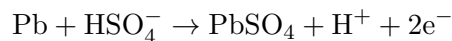
$$V = E_{\text{OCV}} - \eta_{\text{act}} - \eta_{\text{ohmic}} - \eta_{\text{conc}}$$

- At **zero current density**, the cell voltage is approximately the open-circuit voltage.
- At **low current density**, activation polarization dominates, resulting in a rapid voltage drop.
- At **moderate current density**, ohmic polarization dominates, leading to a linear voltage drop.
- At **high current density**, concentration polarization dominates, causing a steep voltage drop.

B4. Consider a lead (anode) and lead-acid (cathode) with H_2SO_4 as the electrolyte. Describe the reactions at the anode and cathode.

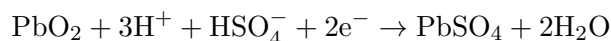
Answer:

Anode oxidation reaction:



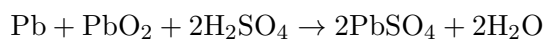
Lead loses electrons. Lead sulfate forms on the anode surface.

Cathode reduction reaction:



Lead dioxide accepts electrons. Lead sulfate and water are formed at the cathode.

Overall reaction:



B5. What is Nernst's equation and what are key parameters?

Answer:

The Nernst equation relates the cell potential to the standard cell potential, temperature, and the activities of reactants and products:

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{n_e F} \ln Q_r.$$

The key parameters are:

- E_{cell} : cell potential under actual conditions.
- E_{cell}^0 : standard cell potential.
- R : gas constant, $8.314 \text{ J mol}^{-1}\text{K}^{-1}$.
- T : absolute temperature in K.
- n_e : number of electrons transferred in the balanced reaction.
- F : Faraday constant, approximately 96485 C mol^{-1} .
- Q_r : reaction quotient, determined by the activities of products and reactants and their stoichiometric coefficients.

B6. How is the Nernst equation applied to batteries? How can it be used to calculate the open-circuit voltage of a cell?

Answer:

The theoretical cell voltage is determined by the Nernst equation. Under no-load conditions, the cell has an open-circuit voltage (OCV) close to this theoretical value. To estimate the OCV, calculate Q_r from the reactant and product activities, then substitute Q_r , T , E_{cell}^0 , and n_e into the Nernst equation.

B7. What is the Butler–Volmer equation and what are the key parameters?

Answer:

The Butler–Volmer equation describes the relationship between electrode current density and overpotential:

$$i = i_0 \left[\exp \left(\frac{\alpha n_e F \eta}{RT} \right) - \exp \left(-\frac{(1 - \alpha) n_e F \eta}{RT} \right) \right].$$

The key parameters are:

- i : net current density.
- i_0 : exchange current density.
- η : overpotential, the deviation of the electrode potential from its equilibrium value.
- α : charge transfer coefficient.

B8. What is Faraday’s law telling us?

Answer:

The amount of chemical change at an electrode is proportional to the electric charge passed. For the same charge passed through different substances, the masses produced or consumed are proportional to their equivalent weights.

B9. In electrified vehicles the following abbreviations are used, i.e., EV, PHEV and HEV. What do these stand for?

Answer:

- **EV**: Electric Vehicle.
- **PHEV**: Plug-in Hybrid Electric Vehicle.
- **HEV**: Hybrid Electric Vehicle.

B10. Describe the principles of Li-ion battery.

Answer:

A Li-ion battery is an electrochemical device that converts stored chemical energy directly into electricity. It operates through the reversible insertion and extraction of lithium ions in the electrode materials.

During charging, an external voltage source drives electrons from the cathode to the anode through the external circuit, while Li-ions move in the same direction through the liquid electrolyte.

During discharge, these processes are reversed. Li-ions move from the anode to the cathode through the electrolyte, while electrons flow through the external circuit to supply power.

The cathode material limits the performance of current Li-ion batteries.

B11. What is the relationship between the electromotive force of a battery and the Gibbs free energy change?

Answer:

The reversible electromotive force of a battery is related to the Gibbs free-energy change of the reaction by

$$\Delta G = -n_e F E, \quad E = -\frac{\Delta G}{n_e F}.$$

Here:

- ΔG : Gibbs free-energy change of the reaction.
- n_e : number of electrons transferred in the reaction.
- F : Faraday constant.
- E : reversible electromotive force, not the actual terminal voltage under load.

For example, a spontaneous discharge reaction has $\Delta G < 0$ and therefore a positive electromotive force.

B12. What are the theoretical capacity, actual capacity, and rated capacity of a battery?

Answer:

- **Theoretical capacity:** the theoretical charge calculated from the active materials and the electron transfer in the reaction, using Faraday's law.
- **Actual capacity:** the charge a battery can deliver under actual discharge conditions. It depends on the discharge rate, temperature, and cut-off voltage.
- **Rated capacity:** the stated ampere-hour capacity of a battery, used as a reference for C-rate and SOC. The actual capacity delivered depends on the discharge conditions.

B13. Why does a battery generate heat during operation?

Answer:

Not all chemical energy is converted into electrical energy. Internal resistance, polarization losses, and reaction entropy changes all contribute to the thermal behavior of a battery.

The main heat sources are:

- **Joule heat:** irreversible heating due to internal resistance, with a heat generation rate of I^2R .
- **Polarization heat:** heating associated with voltage losses due to activation and concentration polarization at the electrode interfaces.
- **Entropy heat:** the reversible heat generated or consumed because of entropy changes during the electrochemical reactions.

B14. What is the mechanism of thermal runaway in batteries?

Answer:

Thermal runaway is a self-reinforcing positive feedback process. When internal heat generation exceeds the battery's ability to dissipate heat, the cell temperature rises. This accelerates exothermic reactions, which generate more heat and cause a further temperature rise.

It can be triggered by overcharge, overdischarge, overheating, external short circuits, or internal short circuits caused by a crash, dendrite formation, or manufacturing impurities. Severe thermal runaway may cause fire or explosion, and failure of one cell can lead to failure of the whole battery pack.

B15. What are the common thermal management system designs for batteries? What are their suitable applications?

Answer:

Battery thermal management systems are classified as passive or active.

- **Passive systems:** natural convection and phase change materials (PCM).
- **Active systems:** forced-air cooling, forced-liquid cooling (e.g., water/glycol), and refrigerant cooling.

Natural convection suits low charge and discharge rates or systems with sufficient space. Forced-air cooling is a simple, low-cost option for moderate heat loads. Liquid cooling suits higher-performance electric vehicles. PCM absorbs heat near its phase-change temperature.

B16. Considering the following three application areas: portable electronics, electric vehicles, and grid-scale energy storage, how do the requirements for energy density, power density, cycle life, and safety differ?

Answer:

The three applications have different priorities, which can be compared in terms of energy density, power density, cycle life, and safety.

Criterion	Portable electronics	Electric vehicles	Grid-scale storage
Energy density	High priority for low weight, compact size, and long runtime.	High priority for driving range within weight and space limits.	Relatively less constrained by weight and volume.
Power density	Device-dependent; laptops and music players require relatively low power.	High power is needed for acceleration and hill climbing.	Depends on the service: sustained peak shaving or rapid response for grid regulation.
Cycle life	Capacity should remain stable over repeated use.	Long life with sustained capacity and power is important.	Long cycle life for frequent operation; less critical for occasional backup.
Safety	Overheating and material decomposition must be controlled.	Particular concern about overcharge, overheating, and thermal runaway.	Large systems require careful control of thermal runaway and flammable-electrolyte risks.

B17. What are the main ageing mechanisms of batteries? How can battery life be extended?

Answer:

For Li-ion batteries, the main ageing mechanisms include:

- Growth of the SEI layer on the anode.
- Loss of cathode material.
- Loss of conductive paths.
- Plating of metallic lithium.
- Electrolyte oxidation.

These processes are associated with capacity and power fade and changes in internal resistance.

Battery life can be extended by:

- Maintaining a suitable, uniform temperature and limiting local overheating.
- Avoiding overcharge and overdischarge and properly limiting charging voltage and current.
- Avoiding excessively fast charging at low temperatures and providing heating when needed.
- Selecting an appropriate SOC range and depth of discharge for the battery type. Partial charging and discharging can extend the life of Li-ion batteries.

Fuel Cells

F1. Is a fuel cell a galvanic or electrolytic cell?

Answer:

燃料电池属于原电池。它通过自发的氧化还原反应，将化学能直接转化为电能。

F2. Describe the principals and components of (a) PEMFC fuel cell, and (b) SOFC fuel cell.

Answer:

(a) 质子交换膜燃料电池 PEMFC

组成: 阳极、阴极、质子交换膜、催化层、气体扩散层和气体流道等。

工作原理: 氢气在阳极催化剂的作用下发生氧化反应，释放电子和质子。质子通过质子交换膜迁移到阴极，而电子通过外部电路流向阴极。在阴极，氧气与质子和电子结合，生成水。

(b) 固体氧化物燃料电池 SOFC

组成: 多孔阳极、多孔阴极、固体氧化物电解质、连接板等，通常在高温下运行。

工作原理: 氧气在阴极接受电子，生成氧离子。氧离子通过电解质到达阳极，与氢气反应生成水并释放电子。电子经外电路流向阴极，向负载供电。

F3. What is meant by enthalpy of formation?

Answer:

生成焓是由组成元素生成某物质时的焓变，通常按生成 1 mol 物质计算。标准生成焓以元素的标准参考状态（25 °C，0.1 MPa）为基准。

F4. What is meant by i_o ?

Answer:

i_o 表示交换电流密度。在电极处于平衡状态时，氧化和还原反应仍同时进行，两个方向的部分电流密度大小相等、方向相反，它们的大小都是 i_o ，而净电流密度为零。

通常 i_o 越大，电极反应越快；在相同电流密度下，所需的活化过电势越小。

F5. The polarization curve of a fuel cell is typically divided into three regions. What physical processes dominate each region?

Answer:

- 低电流密度区：活化极化占主导，电极反应速率限制了电流的增加。
- 中等电流密度区：欧姆极化占主导，主要是电解质和电极材料的电阻导致的电压损失。
- 高电流密度区：浓差极化占主导，反应物消耗速度接近传质供应能力，电极表面反应物浓度下降，导致电压下降。

F6. Describe the polarisation curve for an SOFC. Define efficiency of a fuel cell.

Answer:

SOFC 的极化曲线表示电池电压随电流密度增大的变化：

- 高温下电极反应较快，初始活化电压降较小；开路电压通常接近理论电压。
- 在中等电流密度区，欧姆极化占主导，电压线性下降。
- 在高电流密度区，浓差极化占主导，电压下降加剧。

燃料电池的效率定义：输出电能与输入燃料化学能的比值

$$\eta = \frac{\text{输出电能}}{\text{输入燃料的化学能}}$$

F7. What is the Tafel equation?

Answer:

Tafel 方程描述活化过电势与电流密度对数之间的关系：

$$\eta_{\text{act}} = A \ln \left(\frac{i}{i_0} \right).$$

在每摩尔氢气转移两个电子的情况下，

$$A = \frac{RT}{2\alpha F}.$$

其中， η_{act} 为活化过电势， i 为电流密度， i_0 为交换电流密度， α 为电荷转移系数， R 、 T 、 F 分别为气体常数、绝对温度和法拉第常数。

Tafel 关系对应 Butler-Volmer 方程中一个指数项占主导时的近似。

F8. Starting with the definition of Gibbs energy as $g = h - Ts = u + pv - Ts$ and the first law of thermodynamics $dq = du - p dv$, derive the pressure dependence of Gibbs energy.

Answer:

查阅教材可知，第一定律为 $\delta q = du + p dv$ （题干中的减号应为加号）。对组成固定、仅有体积功的可逆过程，

$$\begin{aligned} g &= u + pv - Ts, \\ \therefore dg &= du + p dv + v dp - T ds - s dT, \\ \therefore \delta q &= du + p dv = T ds, \\ \therefore dg &= v dp - s dT. \end{aligned}$$

恒温时， $dT = 0$ ，因此 $dg = v dp$ 。对理想气体，摩尔体积 $v = RT/p$ ，故

$$g_2 - g_1 = \int_{p_1}^{p_2} \frac{RT}{p} dp = RT \ln \left(\frac{p_2}{p_1} \right).$$

F9. What are the major materials being used in the anode, cathode and electrolyte of a PEMFC?

Answer:

- 阳极和阴极：碳材料与铂催化剂。碳材料导电并支撑铂颗粒，铂催化电化学反应。
- 电解质：质子传导聚合物膜，通常为 Nafion。膜需保持适当水合，以维持质子电导率。

F10. What are the major materials being used in the anode, cathode and electrolyte of an SOFC?

Answer:

- 阳极：镍与氧化钇稳定氧化锆组成的金属陶瓷（Ni-YSZ）。
- 阴极：钙钛矿型氧化物，例如掺杂的锰酸镧 LaMnO_3 。
- 电解质：氧化钇稳定氧化锆（YSZ），即在 ZrO_2 中掺入 Y_2O_3 ，用于传导氧离子。

F11. What is meant by the electric double layer?

Answer:

电双层是电极与电解质接触时，在界面两侧形成的电荷分布。电极表面的电荷吸引电解质中带相反电荷的离子，使界面附近形成两层相互对应的电荷。

电双层是电极反应发生的重要界面区域，其结构会影响电荷转移过程和电极反应速率。

F12. Derive expressions for (a) oxygen consumption rate, (b) hydrogen consumption rate, and (c) water production rate, in a hydrogen–oxygen fuel cell.

Answer:

总反应为 $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ 。根据法拉第定律, 电子摩尔流率为 $\dot{n}_e = I/F$ 。每消耗 1 mol 氧气对应 4 mol 电子, 每消耗 1 mol 氢气或生成 1 mol 水对应 2 mol 电子。因此, 单电池的消耗率和生成率为

$$\dot{n}_{\text{O}_2} = \frac{I}{4F}, \quad \dot{n}_{\text{H}_2} = \frac{I}{2F}, \quad \dot{n}_{\text{H}_2\text{O}} = \frac{I}{2F}.$$

其中, I 为电流, F 为法拉第常数, $\dot{n}$ 的单位为 mol s^{-1} 。对含 N_c 个串联单电池的电池堆, 总速率为上述结果的 N_c 倍。

F13. Describe four types of voltage losses or overpotentials affecting the polarization curve of a fuel cell.

Answer:

1. 活化损失: 电极反应速率有限, 产生活化过电势, 通常在低电流密度区占主导。
2. 欧姆损失: 电解质和电极等材料的电阻造成电压损失, 通常在中等电流密度区占主导。
3. 浓差损失: 反应物消耗接近传质供应能力, 电极表面浓度降低, 通常在高电流密度区占主导。
4. 燃料渗透和内部电流损失: 燃料穿过电解质或电子在内部泄漏, 使部分化学能不能转化为外电路的电功。

F14. What are the major transport processes occurring in a fuel cell?

Answer:

1. 流动与质量传输: 反应气体沿流道流动, 通过对流和扩散到达反应位置, 同时带走产物。
2. 电荷传输: 电子通过电极、连接板和外电路传输, 离子通过电解质传输。
3. 热传输: 通过固体材料导热, 以及流动气体或冷却介质的对流传热。
4. 水传输: 包括扩散、压力梯度引起的对流, 以及 PEMFC 膜中质子迁移引起的电渗拖曳。

F15. What is meant by flooding in a PEMFC?

Answer:

水淹是指生成的液态水不能及时排出, 在电极或气体扩散层的孔隙中积聚, 堵塞气体传输通道, 使反应气体难以到达催化反应位置。这会增加浓差极化并降低电池性能, 尤其影响阴极的氧气供应。因此, 水管理既要防止液态水积聚, 也要保持膜的适当水合。

F16. For a porous material define (a) porosity, (b) tortuosity, and (c) permeability.

Answer:

- **孔隙率 (ϕ):** 孔隙体积与总体积的比值, $\phi = V_{\text{pore}}/V_{\text{total}}$ 。
- **曲折度 (τ):** $\tau = s/L$, 其中 s 为穿过孔隙结构的最短连通路长度, L 为沿传输方向的直线厚度。
- **渗透率 (k):** 表示多孔材料允许流体通过的能力, 单位是 m^2 。在相同流体和压力梯度下, 渗透率越大, 流体越容易通过。

F17. What is Darcy's law?

Answer:

Darcy 定律描述多孔介质中低速流动的表现流速与压力梯度的线性关系。对单相流, 忽略重力和惯性影响时, 一维形式为

$$u = -\frac{k}{\mu} \frac{dp}{dx}.$$

其中, u 为表现流速, k 为渗透率, μ 为动力黏度。负号表示流体沿压力降低的方向流动。

F18. What is the difference between Forchheimer's and Darcy's formulations for a porous media?

Answer:

Darcy 定律忽略惯性影响, 压力梯度与流速呈线性关系:

$$-\frac{dp}{dx} = \frac{\mu}{k} u.$$

流速增大时, Forchheimer 方程增加惯性修正项。按一维正向流动写为

$$-\frac{dp}{dx} = au + bu^2, \quad a = \frac{\mu}{k}.$$

au 为黏性阻力项, bu^2 为惯性阻力项。当惯性项可忽略时, Forchheimer 方程退化为 Darcy 定律。